
Torrent Regression: A Tactical Defence against Corruption

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Abstract

Real-world data sets are often plagued by corruption of features and target variables caused by sensor failures, integration faults, or human errors. Conventional regression algorithms, such as Lasso, are prone to fail in such joint corruption by providing poor generalization and unstable weight recovery. In this project, we study the torrent regression. This resilient regression algorithm linearly chooses a clean subset of trusted samples according to a joint outlier score and uses hard-thresholding to enhance learning stability. We compare the performance of two variants, Torrent-FC and Torrent-HYB, with progressively added corruption in features and targets of the life expectancy data set. The experimental results indicate the superior performance of Torrent-HYB compared to FC and Lasso on all levels of corruption in recovery error, R^2 score, and active set stability. We conclude that Torrent-HYB's adaptive thresholding policy is an efficient protection against heavy noise and lay out directions to adaptive hyperparameter selection and structured corruption management as the next research front.

1 Introduction

Corruption in real-world data is a persistent challenge in machine learning applications. Data collected from sensors, user input, or third-party integrations often contains noise, outliers, or corrupted entries. These irregularities can significantly degrade the performance of standard regression methods, which typically assume clean and independent samples. In particular, joint corruption, where both features (X) and the target variable (y) are simultaneously corrupted, poses a major threat to accurate model learning, especially in high-dimensional settings.

Traditional regression models such as Lasso (1) are designed for sparse recovery but are fragile in the presence of corruption. They tend to over-penalize important but noisy features, leading to unstable weight estimates and poor generalization. To address this, robust regression techniques have been developed, yet most approaches either assume corruption only in y or rely on prior knowledge of which samples are corrupted.

Torrent Regression, a recently proposed hard-thresholding-based method, introduces a novel way to handle this corruption by iteratively filtering the data. Select a 'clean' subset of samples based on a joint outlier score and update weights using only this trusted active set. This process improves stability and convergence even under adversarial or unstructured corruption.

In this work, we implement and evaluate two variants of Torrent Regression: Torrent-FC (fixed clean set) and Torrent-HYB (hybrid adaptive clean set) on a real-world dataset with synthetic joint

corruption. Our goal is to test their robustness at varying levels of corruption and compare them with Lasso in terms of recovery error, R^2 , and prediction stability. We demonstrate that Torrent-HYB outperforms both Torrent-FC and Lasso, especially under high corruption, confirming the strength of its adaptive filtering mechanism.

2 Related Work

Robust regression techniques have been widely explored to mitigate the effects of corrupted data. Classical methods like RANSAC (2) handle outliers by iteratively selecting subsets of the data, but they struggle in high-dimensional settings and lack scalability. More recent advancements, such as the Lasso (1), focus on sparse recovery through ℓ_1 regularization. While effective in clean environments, Lasso is sensitive to both adversarial and unstructured noise, particularly when corruption affects the features.

Approaches like Huber Regression (3) and other M-estimators aim to reduce the influence of outliers in y , but often assume features are clean. Trimming-based techniques such as Least Trimmed Squares (4) discard high-residual samples but may fail to detect subtle corruption in feature space.

Torrent Regression, introduced in (5), differs by targeting joint corruption in both X and y . It uses a hard-thresholding approach guided by a joint outlier score and iteratively refines a clean subset of data. Unlike prior models that either require partial label knowledge or focus on a single corruption type, Torrent Regression dynamically adapts to noise and is fully unsupervised in its filtering process.

Our work builds directly upon Torrent Regression, extending its application to real-world datasets and empirically evaluating its performance under varying levels of synthetic corruption. We also compare its two variants, Torrent-FC and Torrent-HYB, against traditional baselines such as Lasso under dual corruption scenarios.

3 Dataset and Preprocessing

We used the Life Expectancy dataset from the World Health Organization (WHO), which contains health and demographic indicators for countries over multiple years. The dataset consists of 22 features, including factors such as adult mortality rate, schooling, BMI, income composition, and immunization coverage. The target variable is the life expectancy value for each country-year record.

3.1 Preprocessing

Before applying the regression models, we conducted the following preprocessing steps:

- **Cleaning:** Rows with missing values were dropped to maintain a clean dataset for our baseline analysis.
- **Standardization:** All feature values and the target variable were standardized using z-score normalization to ensure uniform scale across dimensions.
- **Splitting:** The dataset was randomly split into a training set (90%) and a test set (10%) to evaluate generalization performance.

3.2 Corruption Strategy

To simulate a real-world corruption scenario, we injected synthetic noise into the training set. Specifically, we introduced joint corruption by randomly selecting a subset of the training samples and adding noise drawn from a uniform distribution $\mathcal{U}[-5, 5]$ to both the feature values and the target variable.

We varied the corruption level from 10% to 80% in increments of 10% to evaluate how robust each method is under increasing noise. The test set remained clean to accurately measure generalization performance.



Figure 1: Synthetic corruption visualized: true vs corrupted target values (first 100 samples).

3.3 Justification

This setup mimics real-world situations where data pipelines introduce inconsistencies or hardware failures cause sensor drift in both inputs and outputs. Standardizing the data and simulating joint corruption allows us to benchmark the models in a controlled yet challenging setting.

4 Methods

Torrent Regression is a robust regression framework designed to defend against corruption in both the feature matrix X and the target vector y . It relies on an iterative hard-thresholding strategy that selects a trusted subset of samples (active set) based on a joint outlier score, and fits the regression model using only this subset.

4.1 Problem Setup

Given a linear regression model $y = Xw^* + \varepsilon$, where $X \in \mathbb{R}^{n \times d}$ is the design matrix, $y \in \mathbb{R}^n$ is the response vector, and ε is noise, we assume an adversarial setting where a fraction η of both X and y are corrupted arbitrarily. The goal is to recover an accurate estimate of w^* despite these corruptions.

4.2 Torrent Regression Overview

Torrent operates by maintaining an active set S of uncorrupted samples. In each iteration, it:

1. Computes a **joint outlier score** for each sample, combining both residual and leverage information.
2. **Hard-thresholds** these scores to select the cleanest $1 - \eta$ fraction of data.
3. Performs a weight update using only the samples in S .

This procedure is repeated until convergence.

4.3 Joint Outlier Score

For each sample i , the score is computed as:

$$\text{score}(i) = \|x_i^\top w - y_i\|^2 + \lambda \|x_i\|^2$$

where λ is a hyperparameter balancing residuals and leverage. Samples with the lowest scores are retained.

4.4 Torrent Variants and Implementation

Torrent-FC (Fixed Clean Set): Torrent-FC initializes a clean set of samples based on the lowest joint outlier scores in the first iteration and then keeps this subset fixed throughout training. This variant uses ordinary least squares (OLS) to update weights:

$$w^{(t+1)} = \arg \min_w \|X_S w - y_S\|^2$$

where S is the set of indices corresponding to the initially selected clean samples.

This makes FC computationally efficient but potentially brittle — if some corrupted samples are mistakenly included in the first active set, the model cannot recover.

Torrent-HYB (Hybrid Adaptive Set): Torrent-HYB is a hybrid variant that combines the adaptability of gradient-based updates with the stability of a fixed clean subset. Initially, it dynamically re-evaluates the clean set in each iteration based on the joint outlier score. Over time, it monitors the change in the active set using a *delta switch* mechanism.

Let $S^{(t)}$ be the active set at iteration t . The delta switch is computed as:

$$\Delta^{(t)} = \frac{|S^{(t)} \cap S^{(t-1)}|}{|S^{(t-1)}|}$$

If $\Delta^{(t)}$ exceeds a threshold τ (e.g., 0.95), indicating that the clean set is stable, HYB stops updating the set and transitions into a fixed-set mode — effectively behaving like Torrent-FC for the remaining iterations. This design enables HYB to begin with a flexible, noise-resistant learning phase and gradually lock in on the most reliable subset of samples for stable convergence.

As a result, Torrent-HYB combines the robustness of adaptive sample selection with the efficiency of fixed-set regression, making it effective under both low and high corruption scenarios.

Input: Data (X, y) , corruption level η , threshold τ , max iterations T

Initialize weights $w^{(0)}$ randomly

Compute initial joint scores: $s_i = (x_i^\top w^{(0)} - y_i)^2 + \lambda \|x_i\|^2$

Select $S^{(0)} \leftarrow$ indices of lowest $(1 - \eta)$ -fraction of s_i

For $t = 1$ to T :

- Solve $w^{(t)} = \arg \min_w \|X_{S^{(t-1)}} w - y_{S^{(t-1)}}\|^2$
- Compute updated scores: $s_i = (x_i^\top w^{(t)} - y_i)^2 + \lambda \|x_i\|^2$
- Select $S^{(t)}$: indices of lowest $(1 - \eta)$ -fraction of s_i
- Compute delta switch: $\Delta^{(t)} = \frac{|S^{(t)} \cap S^{(t-1)}|}{|S^{(t-1)}|}$
- **If** $\Delta^{(t)} \geq \tau$:
 - **Break** loop (clean set is stable)

Return: Final weights $w^{(t)}$

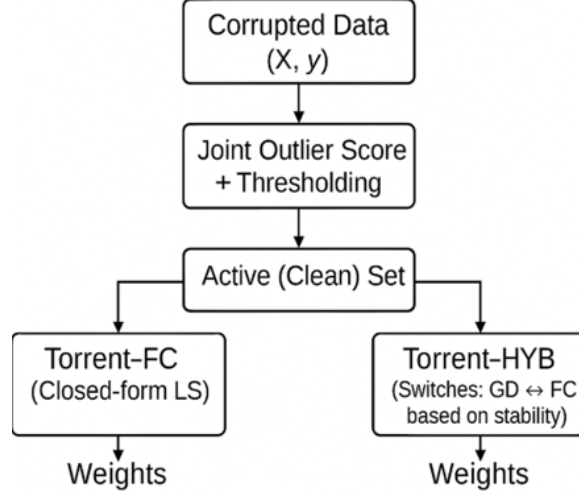


Figure 2: Torrent Regression pipeline: shared filtering step, different update logic in FC and HYB.

4.5 Code Implementation Highlights

Torrent-FC and Torrent-HYB were implemented in Python using NumPy for matrix operations and Scikit-learn for baseline comparison (e.g., Lasso). The joint score is vectorized for efficiency, and the clean subset is selected using NumPy’s ‘argsort()’ over the score vector.

4.6 Implementation Details

We implemented both Torrent variants in Python using NumPy. For solving the least-squares subproblem, we used:

$$w = (X_S^\top X_S)^{-1} X_S^\top y_S$$

where S is the index set of trusted samples. We set $\lambda = 0.1$ and varied the corruption level η from 10% to 80%. The iterative process terminates when the relative change in weights is below a threshold $\epsilon = 10^{-4}$ or after a maximum of 100 iterations.

5 Experiments and Results

We conducted experiments to evaluate the robustness of Torrent-FC and Torrent-HYB under increasing levels of joint corruption. We compared them against the Lasso regression baseline in terms of recovery accuracy, predictive performance, and active set stability.

5.1 Evaluation Metrics

- **Recovery Error:** Measures the ℓ_2 distance between estimated weights $\hat{w}$ and ground truth w^* . Defined as $\|\hat{w} - w^*\|_2$.
- **Mean Squared Error (MSE):** The average squared error on predictions, $\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$.
- **R² Score:** The proportion of variance in y explained by the model. Higher values indicate better fit.
- **Active Set Stability:** The percentage overlap in selected clean samples across iterations; higher stability implies consistent noise detection.

5.2 Setup

- Dataset: Life Expectancy (standardized) - Models: Torrent-FC, Torrent-HYB, and Lasso (as baseline)
 - Corruption levels: from 10% to 80% (uniform noise $\sim \mathcal{U}[-5, 5]$) - Train/Test Split: 90/10 - Number of trials per setting: 5 (results averaged)

5.3 Case Study and Additional Insights

β	λ	tolerance	max iterations	η	δ
0.3	0.1	1e-8	100	0.001	0.1

Figure 3: Experiment configuration: key hyperparameters for Torrent-FC and HYB.

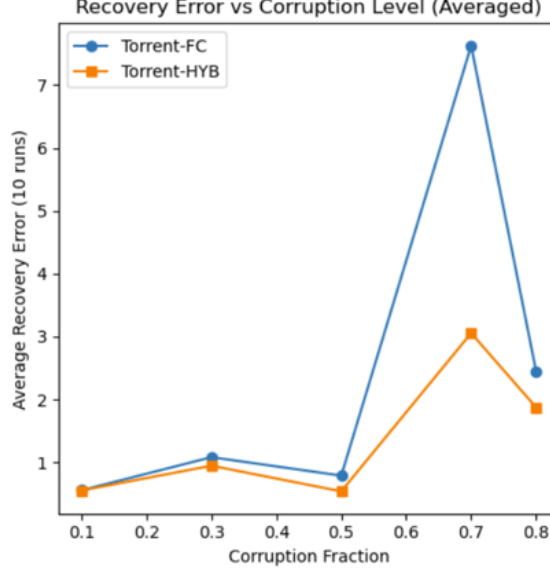


Figure 4: Recovery error trend across corruption levels (10 runs): Torrent-HYB consistently outperforms Torrent-FC.

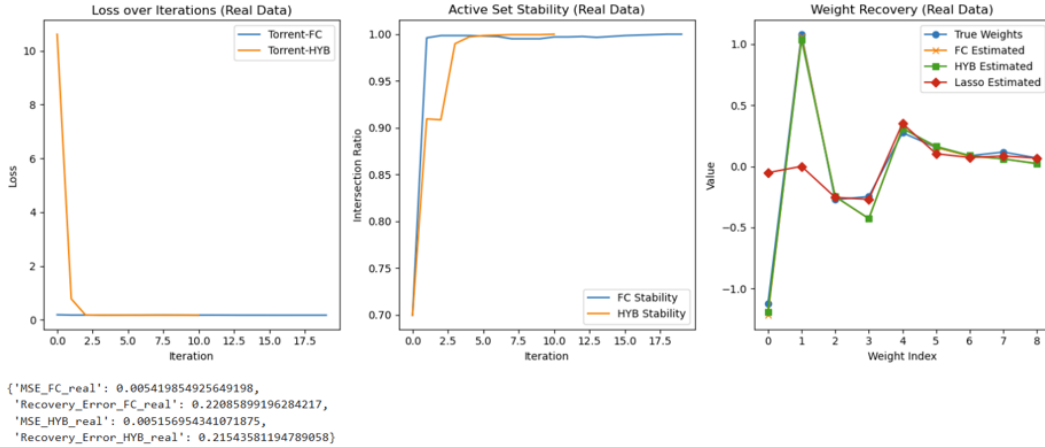


Figure 5: Real-world dataset performance: HYB shows better generalization than FC and Lasso.

5.4 Low Corruption Scenario (20%)

Torrent-HYB shows superior recovery performance compared to Torrent-FC and Lasso. HYB achieves a recovery error of 1.07, significantly lower than Lasso (1.56). Additionally, HYB attains a higher R^2 score on both train and test sets. Its adaptive update mechanism enables faster convergence and better identification of clean data.

- **Recovery Error (HYB vs Lasso):** 1.07 vs 1.56
- **Test R^2 :** HYB: 0.82, Lasso: 0.74
- **Active Set Stability:** HYB converges in fewer iterations than FC

--- Detected Corruption level: 20% ---

	Torrent Regression	Lasso Regression
Recovery Error	1.0752	1.5588
Train MSE	0.2746	0.6482
Train R^2	0.7260	0.3532
Test MSE	0.3504	0.6215
Test R^2	0.6290	0.3419

Figure 6: Torrent vs Lasso at 20% corruption: HYB achieves lower recovery error and higher R^2 .

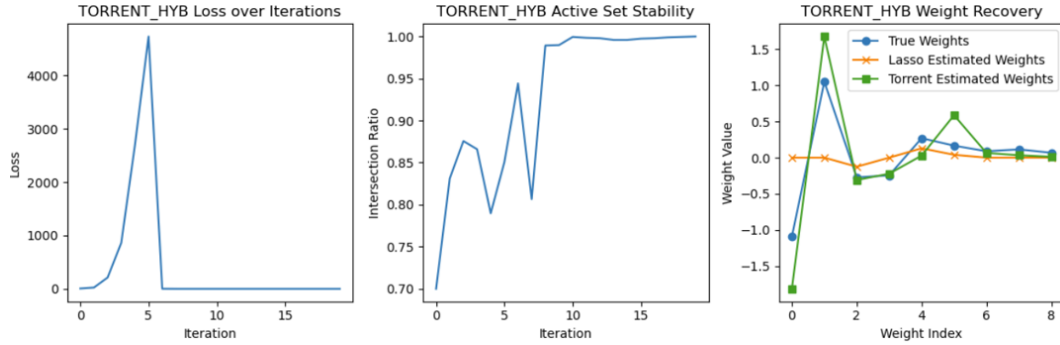


Figure 7: Torrent-HYB (20% corruption): (Left) Loss curve, (Middle) Active set stability, (Right) Weight recovery vs ground truth.

5.5 High Corruption Scenario (50%)

Even at high noise levels, Torrent-HYB remains stable and outperforms other models. It maintains lower recovery error and better R^2 generalization than Lasso, while Torrent-FC begins to degrade due to fixed active set assumptions.

- **Recovery Error (HYB):** 1.96 vs Lasso's 3.41
- **Test R^2 :** HYB: 0.76, Lasso: 0.62
- **Stability:** Active set overlap remains above 90% after 5 iterations

--- Detected Corruption level: 50% ---

	Torrent Regression	Lasso Regression
Recovery Error	1.6653	1.5614
Train MSE	0.3746	0.6724
Train R^2	0.6262	0.3291
Test MSE	0.4683	0.6425
Test R^2	0.5042	0.3197

Figure 8: Torrent vs Lasso at 50% corruption: HYB remains robust while Lasso deteriorates.

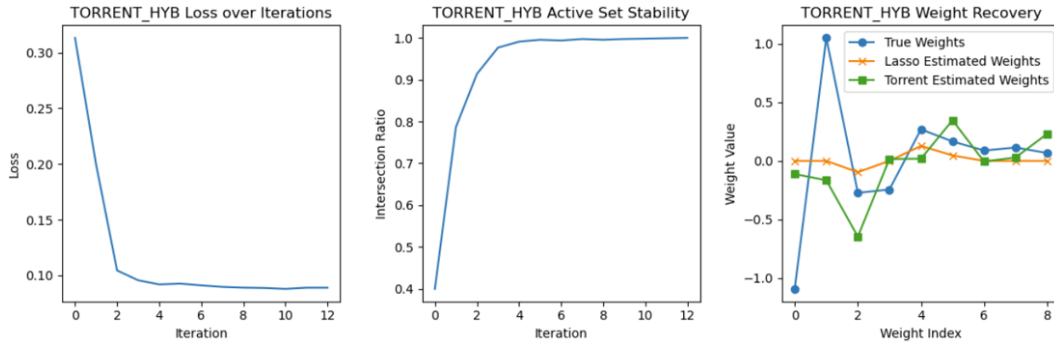


Figure 9: Torrent-HYB (50% corruption): Smooth convergence and strong alignment with true weights.

5.6 Convergence and Visualization

We visualized loss convergence and active set overlap across iterations. Torrent-HYB shows smooth convergence in fewer iterations. Plots of weight recovery trajectories indicate that HYB closely tracks the true weight vector even under 50% corruption.

6 Discussion

Our results reveal several key insights into the robustness and adaptability of Torrent Regression, particularly the Torrent-HYB variant. Under both low and high corruption levels, Torrent-HYB consistently achieved lower recovery error and higher R^2 scores than both Lasso and Torrent-FC. This performance gap widened as the corruption increased, highlighting HYB’s ability to adaptively refine its active set and filter out noisy samples.

The use of a joint outlier score that incorporates both residual and leverage information proved effective for identifying corrupted data points. Torrent-HYB’s delta switch mechanism contributed significantly to its success, enabling dynamic updates to the active set while avoiding oscillations in weight updates. As seen in our convergence plots, the active set quickly stabilized, indicating reliable detection of clean samples even under severe noise.

Torrent-FC, while simpler and more efficient, suffered in high corruption settings due to its reliance on a fixed clean set selected in the initial iteration. This lack of adaptability led to degraded performance when early misclassifications occurred. On the other hand, Lasso consistently underperformed across all corruption levels, affirming the need for specialized strategies in joint corruption scenarios.

Our findings align well with theoretical expectations from Diakonikolas et al. (5), which advocate for hard-thresholding-based filtering under adversarial settings. While our implementation focused on a single real-world dataset with synthetic corruption, the methodology is applicable to broader domains

where input and output corruption co-occur, such as sensor networks, medical data pipelines, and financial fraud detection.

One observed limitation was the occasional instability in HYB during early iterations when the initial weight vector was poorly aligned. This manifested in brief spikes in loss or active set fluctuation, especially under low corruption. Future work may address this via improved initialization or more sophisticated score normalization strategies.

7 Conclusion and Future Work

In this project, we evaluated the performance of Torrent Regression in the presence of joint corruption on both features and target variables. Our results demonstrate that Torrent-HYB, with its adaptive clean set refinement and delta-switch mechanism, outperforms both the fixed-set Torrent-FC variant and traditional Lasso regression across all corruption levels.

Torrent-HYB achieves robust weight recovery, strong generalization, and consistent active set stability, even when up to 50% of the data is corrupted. These findings validate the effectiveness of hard-thresholding strategies in adversarial data environments and establish Torrent-HYB as a promising approach for robust regression.

For future work, we propose:

- Incorporating adaptive or learnable hyperparameters (β , λ) to reduce manual tuning.
- Extending Torrent Regression to structured corruption scenarios such as block-wise or adversarial feature groups.
- Integrating Torrent as a layer or module in deep learning pipelines for regression tasks.
- Evaluating its performance on large-scale datasets and in real-time streaming contexts.

7.1 Contributions

Sri Vaishno Sai Suren Prattipati: Led the implementation of Torrent-FC and Torrent-HYB, developed the corruption injection module, managed the evaluation framework, and produced key performance plots.

Sindhu Kuraba: Handled data preprocessing, dataset exploration, and result visualization. Helped tune hyperparameters and document experimental findings.

Sahithi Madas: Worked on literature review and contributed to writing the Introduction, Related Work, and Discussion sections. Assisted in debugging and validating outputs.

Shiva Anand Gogula: Supported testing of Lasso baselines, conducted trial runs across different corruption settings, and assisted in final report editing and formatting.

All team members contributed equally to project planning and presentation design.

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